

Boris Khesin

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EDUCATION

- **Grinnell College** Grinnell, Iowa
 - *BA in Economics & Computer Science; GPA: 3.77; Dean's List* August 2024 – May 2028 (expected)
 - **Economics & Statistics:** Econometrics (ECN-286), Microeconomic Analysis (ECN-280), International Economics (ECN-233), Applied Statistics (STA-209), Calculus II (MAT-133), Macroeconomic Analysis (ECN-282, Fall 2026)
 - **Computer Science:** Computer Organization & Architecture (CSC-211), Data Structures & Algorithms (CSC-207), Discrete Structures (CSC-208), Software Design & Development (CSC-324), Analysis of Algorithms (CSC-301, Fall 2026), Operating Systems & Parallel Algorithms (CSC-213, Fall 2026)
- **IFSA — Shanghai** Shanghai, China
 - *International Business; Mandarin Chinese* Spring 2027 (expected)
- **Moscow State School 57** Moscow, Russia
 - *Mathematics & Computer Science track; GPA: 4.0* September 2020 – June 2024

WORK EXPERIENCE

- **Topcon Consulting (client: Watson-Forsberg)** Minneapolis, Minnesota
 - *Technology Consulting Intern* June 2026 – July 2026
 - **Client Delivery:** Gathered requirements directly from staff at Watson-Forsberg, a 20-person Minneapolis construction firm, and delivered the internal event-planning system that replaced the three-year-old spreadsheet the firm ran on; in production today across the whole firm, carrying \$40,000 of event budget, 50+ events, a 300+ person directory, and 178 companies.
 - **Data Analysis:** Analyzed event spending across 50+ company events covering hundreds of guests and dozens of clients; consolidated the figures into budget projections and cost breakdowns that informed event planning and vendor selection.
 - **Software Development:** Built the application in Django, integrating it with Unanet, Procore, and the firm's Microsoft systems and adding budget projections and Google Maps; rebuilt the datetime layer under live production data after finding the firm ran multi-day and cross-timezone events the original model did not support.
 - **IT Support:** Served as first-line technology contact for 20 office and construction staff, resolving access and identity questions across Unanet, Procore, and Microsoft Entra and advising end users on the firm's core enterprise platforms.
- **Grinnell College Economics Department** Grinnell, Iowa
 - *Peer Mentor, International Economics (ECN-233)* January 2026 – May 2026
 - **Economics Instruction:** Guided 20+ students through trade theory, international finance, and economic policy; led weekly review sessions analyzing exchange rates, trade agreements, and globalization with real-world data, and gave analytical feedback on quantitative problem sets.
- **Grinnell College Statistics Department** Grinnell, Iowa
 - *Peer Mentor, Applied Statistics (STA-209)* January 2026 – May 2026
 - **Quantitative Methods:** Led weekly sessions on regression, hypothesis testing, and probability using R and Excel for 25–30 students, translating technical concepts across a range of quantitative backgrounds; mentored in two departments in the same semester.
- **Grinnell College Computer Science Department** Grinnell, Iowa
 - *Grader, Functional Programming (CSC-151)* September 2025 – December 2025
 - **Evaluation & Feedback:** Evaluated assignments and exams for a Scheme/Racket functional programming course, writing feedback on code correctness and algorithm design; handled confidential grade records and ran the grading workflow independently to weekly deadlines.
- **Yandex** Moscow, Russia
 - *Mentor/Tutor, Yandex Academy* December 2023 – August 2024
 - **Technical Mentoring:** Became an unpaid mentor/tutor after completing Yandex Academy's backend developer course as a student; tutored 50+ students in C++ and Python one-on-one and in groups of 3–5, reinforcing core programming concepts and supporting students' Django projects through code review.

- **Tinkoff Bank** Moscow, Russia
Product Manager Intern *April 2024 – May 2024*
 - **Central University App:** Worked directly with the product manager on an 8-person team at one of the world's largest digital banks, building a React.js Q&A and gamification app for Central University students; created Figma prototypes and user flows and ran sprint work in Trello.
- **Grinnell College Swimming Team** Grinnell, Iowa
Event Worker *December 2024 – May 2025*
 - **Lane Timer:** Recorded backup times and verified results at competitive meets, including official NCAA events; documented race results to support fair competition and official scoring.

RESEARCH

- **State Corporate Income Taxes and New Business Formation in the United States**
Course research paper (with B. Pilling), Grinnell College *May 2026*
 - **Overview:** Built a balanced state-year panel (51 jurisdictions, 2004–2020; 867 observations) from Census, Tax Foundation, and BEA sources and estimated pooled OLS, two-way fixed-effects, and 2SLS specifications in Stata with state-clustered standard errors. Ran large-state exclusion and a one-year-lead placebo test; the negative cross-sectional association does not survive fixed effects or IV, and the null result is reported.
- **Economic, Demographic, and Policy Factors in COVID-19 Mortality Across Countries**
Course research paper (with B. Pilling and J. Pada), Grinnell College *December 2025*
 - **Overview:** Analyzed COVID-19 deaths per capita and per case across 150 countries in R using Jiangsu University, CIA World Factbook, and World Bank data; fit univariable and multivariable regressions with nested-model F-tests, log transformations, and residual diagnostics, identifying HDI and population density as the strongest respective predictors.

PROJECTS

- **GPT Fine-Tune — Self-Taught Transformers (2023):** Learned transformer architecture from scratch in roughly three weeks with no course or mentor, then fine-tuned a GPT language model on years of cleaned group-chat exports to reproduce how friends text. Engineered compute access independently on an old laptop after finding the usual cloud training services unavailable where I lived. Presented the work at the AI conference of the SISTEMA bootcamp, Moscow Oblast, which led to the product internship at Tinkoff Bank.
- **PIPS — 16-bit CPU & Assembler:** Designed and implemented a microcoded 16-bit CPU in Logisim with a partner for Computer Organization & Architecture (CSC-211): built the datapath with custom ALU, register-file, and shifter subcircuits; wrote the control-ROM microprogram driving 16 opcodes; defined Python assembler rules for 40+ instructions and pseudo-instructions; verified the processor with 14 assembly test programs spanning recursion, arithmetic, and memory-mapped keyboard/terminal I/O.
- **Event Planner (Django):** Built the internal event-planning web app now in production at Watson-Forsberg; synchronized with Unanet, Procure, and Microsoft corporate systems, with budget projections and Google Maps integration. Carries \$40,000 of event budget, 50+ events, 300+ contacts, and 178 companies. (github.com/Shagulka/Event-Planner)
- **Tarot App (iOS):** Built an iOS app for AI-powered tarot readings in Swift and SwiftUI; integrated Gemini-generated interpretations for multiple card spreads and designed the interface to Apple Human Interface Guidelines. (github.com/Shagulka/Tarot-App)
- **Non-Human Vision (VR/XR):** Created a VR/XR simulation of non-human animal visual systems in Unity and C# with a team, writing custom shaders to replicate species-specific color perception and fields of view.

LEADERSHIP & AWARDS

- **Second of 11 Teams, Pioneer Weekend Entrepreneurial Contest:** Owned the financial model and phased rollout for Grin & Go, a campus network of refrigerated smart vending machines, built with a team of five in 48 hours. Modeled unit economics from sourced costs to a break-even at 40.4% of expected spending (roughly 650 of 1,700 students) and \$110,000 of projected annual profit, and staged the launch in three phases so each phase funded the next. Placed second of 11 teams, with personal recognition from Grinnell College President Anne Harris.
- **Debate Leader, Grinnell College Debate Union (Fall 2025 – Present):** Co-lead a student-run debate society: direct weekly meetings, organize competitive travel and resources, and mentor members on argumentation and debate strategy.

- **AI & Competitive Programming, Centre for Teaching Excellence (2023):** Self-taught transformer architectures in three weeks and fine-tuned a GPT language model, engineering compute access on limited hardware; presented at the AI conference of the SISTEMA bootcamp, Moscow Oblast, which led to the Tinkoff internship. Algorithms training: automata, MST, dynamic programming, graphs, strings, segment trees.

SKILLS

- **Quantitative:** Stata, R, Python (pandas, NumPy), Excel; regression, hypothesis testing, panel econometrics (fixed effects, instrumental variables, 2SLS), clustered standard errors, robustness and placebo testing
- **Languages:** Python, C++, Swift, R, SQL, JavaScript, HTML/CSS, Scheme/Racket
- **Frameworks & Tools:** Django, React.js, SwiftUI, Unity (C#), WebGL; Git, LaTeX, Logisim, Figma, Trello; Unanet, Procore, Microsoft Entra
- **Languages:** Russian (native), English (fluent), Mandarin Chinese (elementary; coursework Fall 2026, study abroad Spring 2027), German (beginner)